

A 70-year-old man is scheduled for laryngeal microsurgery for evaluation of hoarseness of voice due to a vocal cord polyp. He has a 50 pack-year history of smoking and chronic obstructive pulmonary disease.

A. Compare endotracheal intubation and ventilation via an anaesthetic circuit with tubeless jet ventilation as a means to manage his airway and ventilation during surgery (50%).

Similarities between the two include:

- Provision of positive pressure ventilation.
- Which risks pneumothorax complication in patient with significant smoking/ COPD history.
- Ability to titrate and provide up to FiO₂ 1.0.
- Shared airway with surgeons, which necessitate close communication during different phases of surgery.
- Risk of airway fire if laser is used.
- Hence, requiring FiO₂ to be reduced as low as possible during laser phase.

Compared to tubeless jet ventilation, endotracheal intubation with ventilation via anaesthetic circuit

- Provides secured airway .
- Which carries lower risk of airway contamination during surgical manipulation.
- Obscures surgical access due to presence of ETT in-situ.
- Carries risks of trauma to vocal cord polyp during intubation.
- Allows for greater control of minute ventilation (tidal volume, respiratory rate, PEEP).
- Allows for effective provision of inhalational anaesthetic agents, with less risk of OT contamination.
- Allows for more reliable monitoring of gas flows (Fi/ET O₂, CO₂, AA).
- Which reduces risks of hypoventilation, hypercapnia, atelectasis.
- Use of HME filter with ETT allows for humidification of gases and better heat retention.
- ETT serves as potential source of fuel for airway fire.